

ACCTG 211H — Mechanics Drill Set (Answer Key)

Week 11 — Cost Behavior and Cost-Volume-Profit Analysis

Drill Set

Quick, single-skill reps to build mechanical fluency before the week's first class. Graded for **completion** — work each one by hand, then check it against the key that follows and re-do any you missed.

1. (*High-low method.*) At 10,000 units of activity total cost is \$80,000; at 6,000 units total cost is \$56,000. Compute (a) the variable cost per unit and (b) total fixed cost.
2. (*Contribution margin.*) A product sells for \$40 and has a variable cost of \$25 per unit. Compute (a) the unit contribution margin and (b) the contribution-margin ratio.
3. (*Break-even in units.*) Fixed costs are \$120,000 and the unit contribution margin is \$15. Compute the break-even point in units.
4. (*Break-even in dollars.*) Fixed costs are \$120,000 and the contribution-margin ratio is 37.5%. Compute break-even sales dollars.
5. (*Target profit.*) Using a unit contribution margin of \$15 and fixed costs of \$120,000, compute the number of units required to earn a target operating profit of \$45,000.
6. (*Margin of safety.*) Budgeted sales are \$500,000 and break-even sales are \$320,000. Compute the margin of safety in dollars and as a percentage of sales.
7. (*Contribution-margin income statement.*) A firm sells 11,000 units at \$40 each; variable cost is \$25 per unit; fixed costs are \$120,000. Prepare a contribution-margin income statement (sales, variable costs, contribution margin, fixed costs, operating income).
8. (*Operating leverage.*) Using the contribution margin (\$165,000) and operating income (\$45,000) from Drill 7, compute the degree of operating leverage, and predict the percentage change in operating income if sales rise 10%.

Self-Check Answers

1. **High-low.** Variable cost per unit = $(80,000 - 56,000) \div (10,000 - 6,000) = 24,000 \div 4,000 = \mathbf{\$6.00}$. Fixed cost = $80,000 - 6.00 \times 10,000 = \mathbf{\$20,000}$ (check at low point: $56,000 - 6.00 \times 6,000 = 20,000$).
2. **Contribution margin.** Unit CM = $40 - 25 = \mathbf{\$15}$. CM ratio = $15 \div 40 = \mathbf{37.5\%}$.
3. **Break-even units.** = $120,000 \div 15 = \mathbf{8,000 \text{ units}}$.
4. **Break-even dollars.** = $120,000 \div 0.375 = \mathbf{\$320,000}$.
5. **Target profit.** = $(120,000 + 45,000) \div 15 = 165,000 \div 15 = \mathbf{11,000 \text{ units}}$.
6. **Margin of safety.** = $500,000 - 320,000 = \mathbf{\$180,000}$; as a percentage = $180,000 \div 500,000 = \mathbf{36\%}$.
7. **Contribution-margin income statement.**

Sales ($11,000 \times \$40$)	440,000
Variable costs ($11,000 \times \$25$)	(275,000)
Contribution margin	165,000
Fixed costs	(120,000)
Operating income	45,000

8. **Operating leverage.** Degree of operating leverage = contribution margin $\div$ operating income = $165,000 \div 45,000 = \mathbf{3.67}$. A 10% rise in sales raises operating income by $3.67 \times 10\% = \mathbf{36.7\%}$.